



Associations Between Diffusion Dynamics and Functional Outcome in Acute and Early Subacute Ischemic Stroke

Chengxia Liu¹ · Shun Zhang¹ · Yihao Yao¹ · Changliang Su¹ · Zhenxiong Wang¹ · Minghuan Wang² · Wenzhen Zhu¹

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Abstract

Purpose The current study aimed to investigate the associations between diffusion dynamics of ischemic lesions and clinical functional outcome of acute and early subacute stroke.

Material and Methods A total of 80 patients with first ever infarcts in the territory of the middle cerebral artery underwent multi-b-values diffusion-weighted imaging and diffusion kurtosis imaging. Multiple diffusion parameters were generated in postprocessing using different diffusion models. Long-term functional outcome was evaluated with modified Rankin scale (mRS) at 6 months post-stroke. Good functional outcome was defined as mRS score ≤ 2 and poor functional outcome was defined as mRS score ≥ 3 . Univariate analysis was used to compare the diffusion parameters and clinical features between patients with poor and good functional outcome. Significant parameters were further analyzed for correlations with functional outcome using partial correlation.

Results In univariate analyses, standard-b-values apparent diffusion coefficient (ADC_{st}) ratio and fractional anisotropy (FA) ratio of acute stroke, ADC_{st} ratio and mean kurtosis (MK) ratio of early subacute stroke were statistically different between patients with poor outcome and good outcome ($P < 0.05$). When the potential confounding factor of lesion volume was controlled, only FA ratio of acute stroke, ADC_{st} ratio and MK ratio of early subacute stroke remained correlated with the functional outcome ($P < 0.05$).

Conclusion Diffusion dynamics are correlated with the clinical functional outcome of ischemic stroke. This correlation is independent of the effect of lesion volume and is specific to the time period between symptom onset and imaging. More effort is needed to further investigate the predictive value of diffusion-weighted imaging.

Keywords Diffusion weighted imaging · Apparent diffusion coefficient · Stroke · Functional outcome · Mean kurtosis

Introduction

The predictive value of clinical measurements such as age, baseline neurological status, baseline level of C-reactive protein, and many others, have been assessed for functional outcome of ischemic stroke [1, 2]; however, prediction of stroke outcome based only on clinical measurements yields only modest accuracy [3]. In recent years, there have been studies in attempts to investigate the potentials of imaging characteristics of ischemic lesions to predict stroke outcome [3–6]. Because destroyed tissues cannot be regenerated in the adult brain, one major mechanism of functional recovery would be reactivation of the functionally disturbed but structurally preserved tissues in ischemic lesions [7]. Thus, a primary requirement of any imaging characteristic of ischemia to associate with the functional outcome would be to quantify the extent of ischemic damage and determine the potentially salvageable tissues.

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✉ Wenzhen Zhu
zhuwenzhen8612@163.com

¹ Department of Radiology, Tongji Hospital, Tongji Medical College, Huazhong University of Science and Technology, 1095 Jiefang Avenue, 430030 Wuhan, China

² Department of Neurology, Tongji Hospital, Tongji Medical College, Huazhong University of Science and Technology, 430030 Wuhan, China

Diffusion-weighted imaging (DWI) has been an essential technique for assessment of ischemic stroke. Apparent diffusion coefficient (ADC) could be used to quantify the extent of restricted diffusion of water molecules in ischemic lesions [8]. A reduction of ADC has been attributed to pathophysiological changes post infarction, such as increased permeability of cell membranes, cytotoxic edema and neurite beading [8–10]. Several studies have tried to correlate regional ADC to the clinical functional outcome of ischemic stroke but produced conflicting results [4–6]. More recently, it was found that conventional ADC obtained with standard b value of 1000 s/mm^2 may not be sufficient to depict the diffusion dynamics as the signal decay of water diffusion no longer follows a linear pattern with increasing diffusion weightings (b values) [11]. A simple approach is to fit the signal decay curve of high b values with another mono-exponential model [12]. There are theoretical advantages of high b value ADC in characterizing cerebral ischemia as the restricted diffusion would be accentuated with increasing b values [13]. Another approach is to depict the curve of signal decay with a two-compartment model. Intravoxel incoherent motion (IVIM) assumes that two distinct proton pools exist inside each voxel and molecular diffusion coefficient (D_{slow}) may differentiate pure molecular diffusion from perfusion-related diffusion [11, 14]. The revised diffusion coefficients of high b value ADC and D_{slow} are thought to characterize ischemic lesions better than the conventional ADC [15–17].

More diffusion parameters have been developed to describe higher order diffusion dynamics of water molecules and they could provide added information of tissue microstructure [18]. Fractional anisotropy (FA) measures the anisotropy of water diffusion which may be shaped by restricting barriers such as cell membranes [19]. In ischemic tissues, FA could quantify microstructural integrity as the orientation of restricting barriers, such as white matter tracts may vary due to ischemic injuries [19, 20]. Mean kurtosis (MK) quantifies the deviation of water diffusion from Gaussian distribution and it has been found to correspond to microstructural destruction in brain injuries [21, 22]. In fact, FA and MK of fiber tracts have been investigated for their potential to predict stroke outcome [23–26]; however, these studies mainly focused on the adaptive structural remodeling or Wallerian degeneration of motor fibers in subacute or chronic stages of stroke. To the best of our knowledge, D_{slow} , high b value ADC, FA, and MK of ischemic lesions have not been evaluated for their potential correlations with functional outcome of acute and subacute stroke.

The current study aimed to explore the associations between the diffusion dynamics (standard- b -value ADC, high- b -value ADC, D_{slow} , FA, MK) of ischemic lesions and functional outcome of acute and early subacute ischemic stroke.

Material and Methods

Patients

This study was approved by the local institutional review board and written informed consent was obtained from all patients. The inclusion criteria were: (1) suspected symptoms of ischemic stroke, (2) time interval between symptom onset and imaging within 3 days and (3) infarct observed on conventional DWI within the territory of the middle cerebral artery. The exclusion criteria were: (1) history of stroke, (2) hemorrhagic transformation detected by T1 weighted imaging (T1WI), (3) small lesions with minimal diameter $\leq 0.5 \text{ cm}$, (4) diffusion abnormalities due to non-vascular etiology, (5) severe systemic diseases, (6) motion artifacts or severe distortion on DWI and (7) recurrent stroke during follow-up. In order to allow evaluation of spontaneous functional recovery, patients with systemic intravenous thrombolytic therapy or endovascular treatment were also excluded. All patients were treated with standard clinical care to achieve the maximum potential functional improvement. On admission, patients were evaluated for stroke severity using the National Institutes of Health stroke scale (NIHSS) by a neurologist (5 years of experience). At 6 months after symptom onset, functional outcome was assessed using the modified Rankin Scale (mRS) by another neurologist (4 years of experience) who was blinded to all baseline clinical and imaging data. Functional outcome was dichotomized into good (mRS score ≤ 2) and poor ($3 \leq \text{mRS score} \leq 6$).

Magnetic Resonance Imaging (MRI)

All scans were performed on a 3.0T MRI system (MR 750; GE Healthcare, Waukesha, WI, USA) with a 32-channel receiver head coil. The scan protocol included T1WI, T2-weighted imaging (T2WI), fluid-attenuated inversion recovery (FLAIR), multi- b -value DWI and diffusion kurtosis imaging (DKI), all of which were performed during the same examination for direct comparisons. Multi- b -value DWI was acquired using a spin-echo echo-planar imaging (SE-EPI) diffusion sequence with 19 b values (0, 20, 50, 100, 150, 200, 400, 600, 800, 1000, 1200, 1500, 2000, 2400, 2800, 3200, 3600, 4000, 4500 s/mm^2) in three orthogonal directions. Other scan parameters were: repetition time (TR) 3200 ms; echo time (TE) 90.7 ms; matrix 160×160 ; number of slices 20; slice thickness 5 mm; spacing 1.5 mm and field of view (FOV) $240 \times 240 \text{ mm}^2$. The acquisition time was 5 min and 52 s. The DKI used a SE-EPI sequence with b values of 1250 and 2500 s/mm^2 applied in 25 uniformly distributed directions. The two images of b_0 were acquired and other parameters were as follows: TR 6500 ms; TE 82.3 ms; matrix 128×128 ; number of slices 43; slice thick-

ness 3 mm; spacing 0 mm; and FOV 256 × 256 mm². The acquisition time was 5 min 45 s.

Image Analysis

Data were transferred to a workstation (AW 4.5, GE Healthcare) for post-processing. Metric maps of standard-b-value ADC (ADC_{st}) were calculated with b values of 0 and 1000 s/mm². Metric maps of high-b-value ADC (ADC_{hi}) were obtained with b values ranging from 2000 to 4500 s/mm² based on the algorithm reported previously [12]. Metric maps of D_{slow} were fitted using the IVIM model [11]. Metric maps of MK and FA were generated with the DKI model [21].

Image registration was performed with a semi-automatic technique supplied by the workstation vendor. The transverse T1WI was used as a reference to which all metric maps were registered. The regions of interest (ROIs) in the ischemic lesion were manually identified by two neuroradiologists with conspicuous hyperintensity on the standard-b-value DWI images and were then copied to the contralateral hemisphere. The ROIs were carefully delineated and adjusted to avoid any incursion of fluid from brain ventricles or sulcus. The ROIs on both hemispheres were then propagated to all metric maps and the mean values of all parameters were obtained simultaneously. Measurements obtained by the two neuroradiologists were averaged for each parameter for statistical analysis. The ratios of lesioned/contralateral measurements were computed for all diffusion parameters. Lesion volume was measured by delineating the hyperintensities on each slice of the standard b-value DWI maps where the lesions appeared.

Statistical Analysis

Continuous variables were presented as mean ± standard deviation (SD). Statistical analysis was performed using SPSS software (Version 20.0, IBM, Armonk, NY, USA) with respect to data distribution. Because the diffusion parameters are dynamic throughout the time course of stroke [27–31] and this may affect their potential correlation with the functional outcome, the analyses were carried out in acute stroke (time to imaging within 24 h) and early subacute stroke (time to imaging between 24 h and 72 h) separately. The two-tailed paired t-test or Wilcoxon signed ranks test was used to compare the diffusion parameters between the stroke areas and the contralateral normal areas. In univariate analyses, the diffusion parameters, lesion volume, dominant hemisphere, and a series of clinical characteristics were compared between the group of poor outcome and the group of good outcome using Student's t-test, Mann-Whitney U-test or Fisher's exact test. Because prediction of long-term functional outcome is likely subjected to mul-

tiples factors, only factors that were statistically significant in the univariate analyses were further controlled in the partial correlation analysis between the diffusion parameters and the functional outcome. The interobserver variability of measurements was assessed with intraclass correlation coefficients. Statistical significance was defined at $P < 0.05$.

Results

Patient Characteristics

From February 2013 to December 2014, a total of 91 patients were enrolled into this study, 7 patients with severe motion artifacts and 4 with recurrent stroke during the follow-up were excluded. Finally, 80 patients with baseline and follow-up data were included (62 males, 18 females; mean age 56.3 ± 11.6 years, age range 24–78 years). Table 1 summarizes the baseline clinical characteristics and the general imaging characteristics of the stroke patients in both acute and early subacute stroke.

The metric maps of ADC_{st}, ADC_{hi}, D_{slow}, FA, and MK are shown in Fig. 1. In both acute and subacute patients, all diffusion parameters showed statistically significant differences between the stroke areas and contralateral normal areas (Supplementary Tables 1 and 2 of Online Resources). This is consistent with previous separate studies [15–17, 32–34]. Among the five diffusion parameters studied, the largest percentage change was observed in MK in both acute and early subacute stroke. The percentage changes of MK were slightly larger in the early subacute stage than in the acute stage. The intraclass coefficients were between 0.827 and 0.955, which indicated a generally small interobserver variability in all measurements (Supplementary Table 3 of Online Resources).

Inter-group Comparisons of Clinical Characteristics and General Imaging Characteristics

Table 1 shows that the lesion volume was significantly smaller in the patient group of good outcome than of poor outcome in both acute stroke ($P < 0.01$) and early subacute stroke ($P < 0.001$). The initial stroke severity as evaluated by the admission NIHSS score was significantly milder in the patient group of good outcome than of poor outcome only in subacute stroke ($P < 0.01$) but not in acute stroke ($P = 0.142$). Because the admission NIHSS score is actually a clinical result of the ischemic tissue at baseline, just like the mRS score on follow-up, it was not controlled in the partial correlation analysis in the group of early subacute stroke. Then only the lesion volume was controlled in the correlation analysis between the diffusion parameters and the functional outcome.

Table 1 Baseline clinical characteristics of the patient groups of poor and good outcomes

	Acute			Early subacute		
	Poor outcome (<i>n</i> = 13)	Good outcome (<i>n</i> = 20)	<i>P</i> value	Poor outcome (<i>n</i> = 22)	Good outcome (<i>n</i> = 25)	<i>P</i> -value
Age (years)	57.62 ± 14.77	53.55 ± 13.22	0.302	57.18 ± 11.00	56.92 ± 9.12	0.957
Sex, male/female	10/3	15/5	1.000	15/7	22/3	0.154
Admission NIHSS ^a	6 ± 7	4 ± 10	0.142	8 ± 9	4 ± 3	<0.001
Smoking, <i>n</i> (%)	5 (38)	7 (35)	1.000	12 (55)	14 (56)	1.000
Alcohol, <i>n</i> (%)	5 (38)	8 (40)	1.000	11 (50)	14 (56)	0.773
HBP, <i>n</i> (%)	12 (92)	13 (65)	0.108	16 (73)	16 (64)	0.550
DM, <i>n</i> (%)	4 (31)	3 (15)	0.393	7 (32)	8 (32)	1.000
Dyslipidemia, <i>n</i> (%)	5 (38)	7 (35)	1.000	8 (36)	15 (60)	0.147
Dominant hemisphere, <i>n</i> (%)	4 (31)	4 (20)	0.681	8 (36)	14 (56)	0.244
DWI lesion volume ^a	337.7 ± 860.50	29.26 ± 305.24	<0.01	151.73 ± 654.61	34.96 ± 51.44	<0.001

^a Indicates median ± interquartile range

NIHSS National Institutes of Health stroke scale, HBP high blood pressure, DM diabetes mellitus, DWI diffusion-weighted imaging

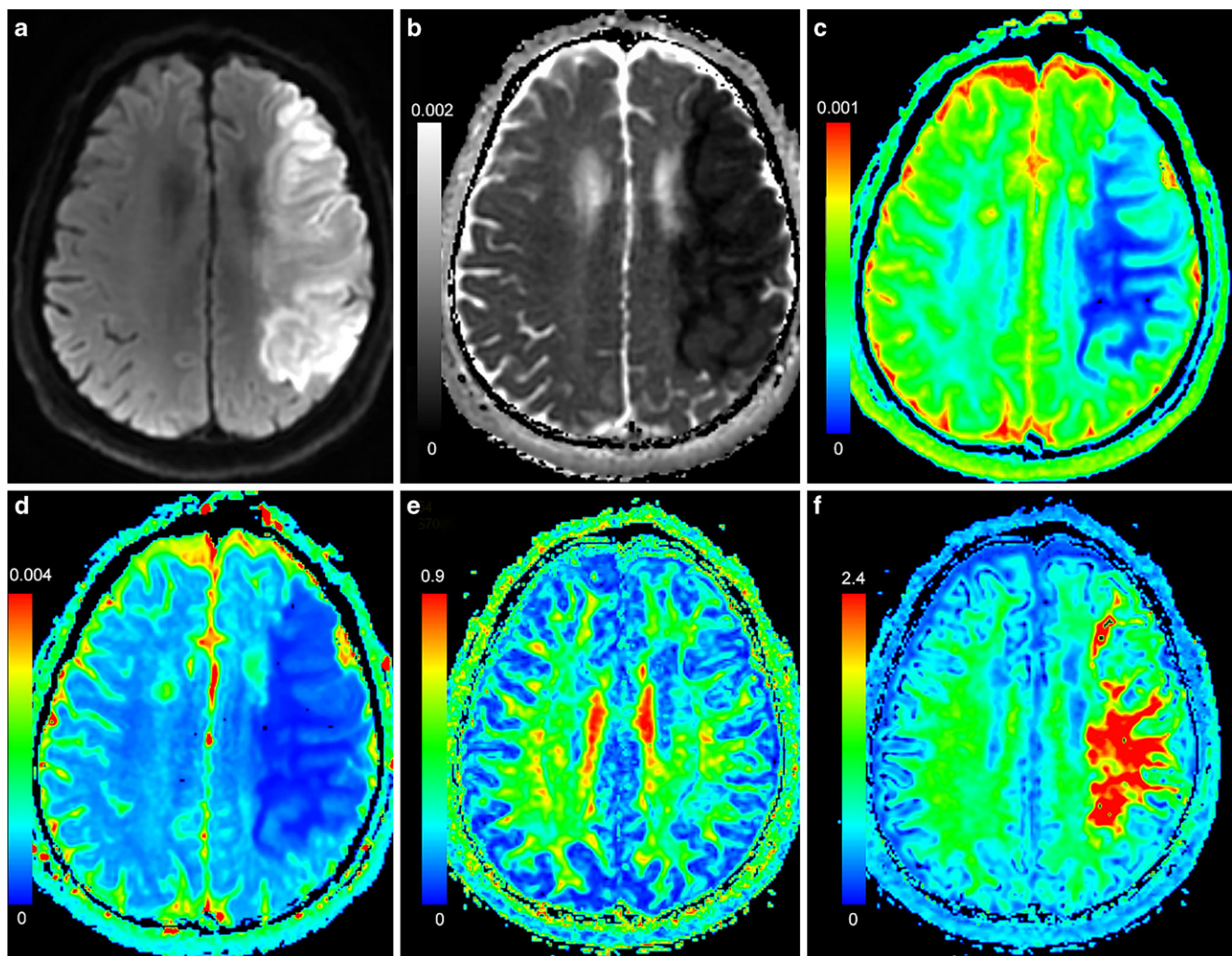
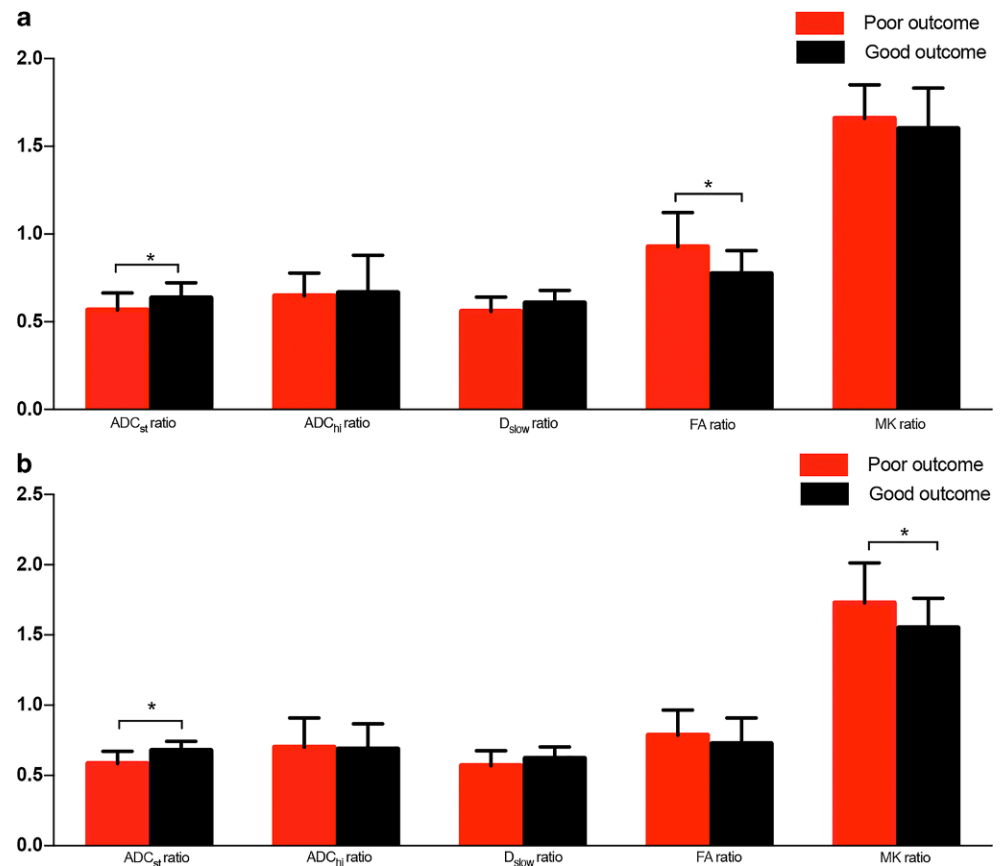
**Fig. 1** Metric maps of a 59-year-old man with acute ischemic stroke. Diffusion weighted image shows the ischemic lesion in the left frontoparietal lobe (a). The extent of the ischemic lesion is generally the same in standard-b-values apparent diffusion coefficient (b), high-b-values apparent diffusion coefficient (c), and molecular diffusion coefficient (d). The ischemic lesion is much smaller in mean kurtosis (e) and vague in fractional anisotropy (f)

Fig. 2 Bar graphs of the diffusion parameters between the group of poor outcome and the group of good outcome in acute stroke (a) and early subacute stroke (b). Error bars indicate standard deviations across subjects. ADC_{st} , ADC_{hi} , and D_{slow} are in units of $10^{-3} \text{ mm}^2/\text{s}$. Asterisk indicates $P < 0.05$. Parameters not marked with asterisks are not significant



Correlation between Functional Outcome and Diffusion Parameters

Figure 2 shows the comparisons of the diffusion parameters between the group of poor outcome and the group of good outcome in acute stroke and early subacute stroke. In acute stroke the ADC_{st} ratio was significantly lower in the patient group of poor outcome than of good outcome whereas the FA ratio was significantly higher in the patient group of poor outcome than of good outcome ($P < 0.05$). In early subacute stroke, ADC_{st} ratio was significantly lower in the patient group of poor outcome than of good outcome while MK ratio was significantly higher in the patient group of poor outcome than of good outcome ($P < 0.05$). No associations of functional outcome with ADC_{hi} ratio and D_{slow} ratio were

observed in either acute and subacute stage. When the lesion volume was controlled, FA ratio of acute stroke, ADC_{st} ratio and MK ratio of early subacute stroke were still moderately correlated with the functional outcome (Table 2).

Discussion

This study was an initial attempt to correlate the diffusion parameters with the functional outcome of ischemic stroke. In the study, regional FA correlated with the functional outcome in acute stroke while ADC_{st} and MK correlated with the functional outcome in early subacute stroke. No significant associations were observed between ADC_{hi} , D_{slow} , and the functional outcome in either acute or subacute stroke.

In two previous studies correlations were found between ADC reduction and the functional outcome [4, 5] in stroke up to 60 h after onset but another study failed to find similar correlations in acute stroke [6]. The present study showed that ADC_{st} correlated with the functional outcome in the early subacute stroke but not in the acute stroke. Together this evidence implies that the association between ADC_{st} and functional outcome is specific to the time intervals between symptom onset and imaging. This is supported by the dynamic changes of water diffusion corresponding to

Table 2 Partial correlation between diffusion parameters and mRS scores with lesion volume controlled

	Parameter	Correlation coefficient	P-value
Acute	ADC_{st} ratio	-0.168	0.359
	FA ratio	0.390	<0.05
Early subacute	ADC_{st} ratio	-0.502	<0.001
	MK ratio	0.345	<0.05

ADC_{st} Standard-b-value apparent diffusion coefficient, FA fractional anisotropy, MK mean kurtosis

the evolvement of underlying pathophysiology post-stroke [27–30]. The most significant drop of ADC ratio occurred in the first day post-stroke and then it decreased slowly until the rise on the fourth day [30]. It is likely that ADC_{st} of early subacute stage endures less effects of the acute stage and better represents the pathophysiological changes of energy metabolism and cell swelling. Alternatively, the association may also be partly explained by the fact that ADC_{st} of early subacute stage was acquired in a time closer to the final destiny of ischemic tissues.

In the current study, no significant association of functional outcome was observed to the revised diffusion coefficients of D_{slow} and ADC_{hi} in either the acute or early subacute stage. This may be associated with the subtle differences of the underlying pathophysiology these diffusion parameters represent. The ADC_{st} may reflect both water diffusion and microcirculation in ischemic tissues [14]. The D_{slow} tries to deplete the effects of tissue perfusion and reveals pure diffusion dynamics of water molecules [11]. Signals of normal tissues may be more effectively suppressed in ADC_{hi} with increasing diffusion weighting [35] but regional microcirculation and microstructure have been important in the evaluation of tissue recovery in ischemia [36]. In this sense, ADC_{st} may be a more comprehensive marker of tissue viability than D_{slow} and thus better associated with functional outcome. In addition, because the signal intensity would decrease exponentially as the diffusion weighting (b values) increases, ADC_{hi} has been found to suffer from a low signal-to-noise ratio which indicates that the diffusion signal is subject to noise contamination [37]. Also, the different temporal dynamics of the diffusion parameters post-stroke [16, 31, 37] and the relatively small study cohort may also account for the insignificant results.

Ischemic injuries of human brain are heterogeneous and MK seems to be more sensitive to severe ischemic damage [33]. This is supported by the observation that MK showed the largest percentage changes between the stroke areas and the contralateral normal areas in our study. In an experimental stroke, ischemic tissues with concurrent diffusion and diffusion kurtosis deficits recovered poorly on reperfusion whereas regions with only diffusion changes responded reasonably well [38]. Furthermore, a Monte Carlo study revealed that kurtosis is specifically sensitive to cell size and membrane permeability but not cell volume fraction [39]. When membrane permeability and cell volume fraction remain fixed, the kurtosis rises up to about 60% when cell size decreases from 10 μm to 5 μm [39]. Then the observed sharp increase of MK in ischemia seems to indicate that microstructural damages might play a more prominent role than cell swelling in ischemic tissues, as the latter would decrease the MK value. The high sensitivity of MK to severe ischemic damages, which may be more prevalent in later stage of stroke, may be attributed

to its correlation with the functional outcome in the early subacute stage, as more severe ischemic damages revealed by rapid MK increase are more likely ensued with poor functional outcome.

In our study, the positive correlation between FA and the mRS score means that the larger the FA ratio value is, the higher the mRS score and the poorer the outcome. This seems to contradict the corticospinal tract studies in which higher FA values were associated with better motor function; however, our study mainly evaluated the extent of damages to the ischemic lesions per se other than the secondary degeneration of the corticospinal tract. Actually, FA drops more dramatically in white matter than in gray matter in ischemia [28] and stroke involving the gray matter tends to have poor functional recovery [40]. Thus, comparing with pure white matter infarctions, stroke with gray matter infarctions have poorer functional outcome and milder FA decrease. In addition, FA value is generally smaller in gray matter than in white matter in a normally aging population [41] and the smaller denominators may further increase the FA ratio in gray matter infarctions.

Prediction of the long-term functional outcome of stroke is very complicated as the outcome is subjected to multiple factors. Our study made the utmost efforts to standardize the baseline characteristics of stroke patients, including enrolling only first ever patients with infarcts within the territory of the middle cerebral artery, performing standardized clinical care, and excluding patients with intravenous treatment or endovascular treatment. Because diffusion parameters are dynamic post-stroke, patients were classified into acute stroke group and early subacute stroke to minimize the potential effects of the time interval between symptom onset and imaging on stroke outcome; however, several limitations exist in the current study. First, we did not evaluate the potential intercorrelation between the clinical characteristics and the imaging parameters although the univariate analyses and partial correlation analyses were used to exclude potential confounders. Second, the effect of lesion distribution on the functional outcome was not evaluated. Third, the mechanisms of the relationship between the diffusion parameters and the functional outcome are generally speculative.

Conclusion

Diffusion dynamics of ischemic lesions are correlated with the clinical functional outcome of ischemic stroke. The correlation is independent of the effect of lesion volume and is specific to the time period between symptom onset and imaging. More effort is needed to further investigate the relationship between DWI and the functional outcome of ischemic stroke.

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Compliance with ethical guidelines

Conflict of interest C. Liu, S. Zhang, Y. Yao, C. Su, Z. Wang, M. Wang and W. Zhu declare that they have no competing interests.

Ethical standards This study was approved by the local institutional review board and written informed consent was obtained from all patients.

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